CHAPTER 3

COMPONENTS OF ROBOTS AS A SYSTEM

What are gears?

Gears are mechanical devices with toothed wheels or cylinders that mesh together to transmit and control motion, torque, and speed between rotating shafts. They are essential to many machines including robots, automobiles, industrial equipments etc.

Uses of Gears

- increase or decrease speed
- amplify torque
- change the direction of rotation

Functions or principles of Gears?

1. Torque Transmission: Gears transmit rotational force (torque) from one shaft to another.
2. Speed Change: Gears can increase or decrease speed, depending on the gear ratio.
3. Direction Change: Gears can change the direction of rotation.
4. Mechanical Advantage: Gears provide mechanical advantage, allowing robots to lift heavier loads or move more efficiently.
4. Types of Gears? (Study diagram for identification from text book. Pgno: 68)

Here are the common types of gears used in robotics:

1. Spur Gears

- Straight teeth, parallel to axis

- High efficiency, low backlash

- Used in robotic arms, conveyor systems

2. Helical Gears

- Angled teeth, non-parallel to axis

- Smooth operation, high load capacity

- Used in robotic joints, transmission systems

3. Bevel Gears

- Teeth on conical surface

- Changes direction of rotation

- Used in robotic arms, grippers

4. Worm Gears

- Screw-like threads

- High gear ratio, compact design

- Used in robotic joints, linear actuators

5. Planetary Gears

- Multiple gears on a common axis

- Compact, high-torque, high-ratio

- Used in robotic arms, mobility systems

6. Rack and Pinion Gears

- Linear motion conversion

- High precision, low backlash

- Used in robotic linear actuators, conveyor systems

7. Cycloidal Gears

- Unique tooth profile

- High-ratio, low-backlash, compact

- Used in robotic joints, precision mechanisms

What is Gear Ratio?

Gear ratio is a critical concept in robotics, affecting the performance, efficiency, and precision of robotic systems. Gear ratio is the ratio of the input speed (or torque) to the output speed (or torque) of a gear system.

Types of Gear Ratios:

1. Speed Increasing Gear Ratio: Increases output speed, reduces torque.

2. Speed Reducing Gear Ratio: Reduces output speed, increases torque.

3. Torque Multiplying Gear Ratio: Increases output torque, reduces speed.

Calculating Gear Ratio:

Gear Ratio (GR) = Number of Teeth on Output Gear / Number of Teeth on Input Gear

GR = Output Speed / Input Speed

Factors Affecting Gear Ratio Selection:

1. Load Capacity: Higher gear ratio for heavier loads.

2. Speed Requirements: Higher gear ratio for faster speeds.

3. Efficiency: Optimize gear ratio for maximum efficiency.

4. Precision: Higher gear ratio for precise control.

5. Space Constraints: Compact gearboxes for smaller robots.

Here are some examples to illustrate gear ratio:

Example 1: Robotic Arm

*Input Gear (Motor): 12 teeth

*Output Gear (Arm): 48 teeth

*Gear Ratio: 48/12 = 4:1

Example 2: Mobility System

*Input Gear (Motor): 20 teeth

*Output Gear (Wheel): 5 teeth

*Gear Ratio: 5/20 = 1:4

Example 3: Gripper Mechanism

*Input Gear (Motor): 15 teeth

*Output Gear (Gripper): 60 teeth

*Gear Ratio: 60/15 = 4:1

Example 4: Bicycle

*Input Gear (Pedals): 52 teeth

*Output Gear (Wheel): 26 teeth

*Gear Ratio: 26/52 = 1:2

What are sensors?

Sensors in robotics are devices that detect and measure various physical or environmental parameters, providing valuable data to enable robots to:

1. Perceive their environment

2. Interact with objects

3. Make decisions

4. Adapt to changes

What are the types of sensors based on application?

- Vision Sensors - Is a type of sensors that captures visual information from the surrounding environment. Examples
  Cameras (visible, infrared, ultraviolet)
  Stereo vision systems
  Structured light sensors
  Lidar (Light Detection and Ranging)
- Tactile and Force Sensors - It is type of sensor that detects and measures physical contact or pressure.
  Example:
  Touch sensors
  Pressure sensors
  Force sensors
  Load cells
- Temperature sensors - It is a type of sensor that measure the temperature of a substance, environment, or object.
  Example:
  Thermocouples (TC)
  Resistance Temperature Detectors (RTD)
  Thermistors
  Infrared (IR) Temperature Sensors
  Digital Temperature Sensors
  Temperature Transmitters
  Pyrometers
- Range and Proximity Detection Sensors (Distance sensors) - It is a type of sensor used to measure the distance between the robot and objects in its environment.
  Example:
  Ultrasonic sensors
  Infrared sensors
  Radar sensors

Lidar sensors

Stereoscopic vision sensors

- Motion and Velocity Sensors - It is a type of sensor that detects movement or changes in the environment.

  Passive infrared sensors

  Optical Encoders

  Tachometers

  Accelerometers

  Gyroscopes

- Navigation and Localization Sensors - It is a type of sensor that is used to provide information about the robot's position, orientation or movement in its environment.

  Examples:

  GPS (Global Positioning System)

  Inertial Measurement Units (IMUs)

  Accelerometers

  Gyroscopes

  Magnetometers

  Odometry sensors

- Speech Recognition Sensors: Speech recognition sensors, also known as speech-to-text (STT) sensors, are devices that convert spoken language into digital text. These sensors use various technologies to detect and interpret sound waves, enabling machines to understand human speech.

  Examples:

  Acoustic Sensors

  Micro-Electro-Mechanical Systems (MEMS) Microphones

  Digital Signal Processing (DSP) Sensors

  Artificial Intelligence (AI) Powered Sensors

  Deep Learning-based Sensors

CLASSIFICATION OF SENSORS (based on their physical location)

- Internal Sensors
- External Sensors

INTERNAL SENSORS

They are placed within the body or structure of the robot. They monitor the robot's internal state and gathers data from within the body itself. These sensors are crucial for maintaining optimal performance, safety and efficiency.

Types of Internal Sensors:

a) **Position sensors**: Measures the position or displacement of an object or component. Example: Potentiometer, Encoders , Hall effects sensors
b) Velocity sensors: Measures the speed or rate of change of position of an object. Examples: Tachometers, Doppler sensors
c) Acceleration sensors: It is also called accelerometers and measures the rate of change of velocity or acceleration of an object. Example: MEMs accelerometer, Gyroscopes
d) Force sensors/ Load sensors: Measures the force or pressure applied to an object. Examples: Strain Gauges, Load cells

EXTERNAL SENSORS

External sensors in robotics detect and measure external stimuli, enabling robots to interact with and respond to their environment.

Types of External Sensors

a) Contact - Type External Sensors
    - Physically touch or contact the object/medium.
    - Measure properties through direct interaction.
    - Typically require physical connection or proximity.
    - Examples: Tactile sensors (force, pressure, touch), Temperature sensors (thermocouples, thermistors)
    -

b) Non- Contact- Type External Sensors

- Do not physically touch or contact the object/medium.
- Measure properties through indirect interaction (radiation, waves).
- Typically require line-of-sight or proximity.

- Examples: Cameras (visual, infrared), Radar sensors (distance, velocity), Lidar sensors (distance, mapping), Ultrasonic sensors (distance, proximity), GPS receivers (location, velocity)

Difference between internal sensors and external sensors

Key Differences:

1. Focus: Internal sensors focus on the robot's internal state, while external sensors focus on the environment.

2. Purpose: Internal sensors monitor robot performance, while external sensors enable interaction with the environment.

3. Data Type: Internal sensors provide proprietary data, while external sensors provide environmental data.

4. Range: Internal sensors have a shorter range, while external sensors can detect objects at longer distances.

5. Accuracy: External sensors require more advanced processing for accurate data interpretation.

FUNCTIONS OF SENSORS

Sensors in robotics enable robots to perceive, interact with and respond to their environment, performing various functions:

Primary Functions:

1. Perception: Gather information about the environment, objects and events.

2. Measurement: Quantify physical parameters (temperature, distance, force).

3. Detection: Identify presence, absence or changes in stimuli.

4. Recognition: Classify objects, patterns or events.

5. Tracking: Monitor movement, position or orientation.

ACTUATORS

Actuators in robotics are components that convert energy into motion, enabling robots to interact with and manipulate their environment.

Types of Actuators

Linear Actuators (translational motion)

Linear actuators produce linear motion in a straight line.

Types of Linear Actuators:

1. Hydraulic Cylinders: They use hydraulic fluid to generate linear motion.

2. Pneumatic Cylinders: They use compressed air or gas to create linear motion

3. Electric Linear Actuators (ELAs) : They use electric motors to create linear motion

Examples:

1. Industrial robots (KUKA, ABB)

2. Autonomous vehicles (Tesla, Waymo)

3. Medical devices (da Vinci Surgical System)

Rotary Actuators (rotational motion)

Rotary actuators provide rotational motion around a fixed axis.

Types of Rotary Actuators:

1. Electric Motors (DC motors, AC motors, Stepper motors, servo motors) : They use electric motors to produce rotational motion.

2. Hydraulic Motors : They use hydraulic fluids to produce rotational motion.

3. Pneumatic Motors : They use compressed air or gas to produce rotational motion.

Examples:

1. Industrial robots (KUKA, ABB)

2. Autonomous vehicles (Tesla, Waymo)

3. Medical devices (da Vinci Surgical System)

Applications of Actuators

Actuators have diverse applications across various industries, enabling precise motion control, automation and interaction with environments.

1) **Industrial Applications:**

1. Robotics (assembly, material handling)

2. CNC Machines (computer numerical control)

3. Automation Systems (conveyor belts, sorting)

4. Manufacturing (production lines, quality control)

5. Packaging and Labeling

6. Material Handling (cranes, hoists)

7. Aerospace (aircraft control surfaces)

8. Automotive (engine control, transmission)

2) Medical Applications:

1. Surgical Robotics (da Vinci, robotic-assisted surgery)

2. Prosthetic Limbs (DEKA Arm System)

3. Exoskeletons (rehabilitation, paralysis support)

4. Medical Imaging (MRI, CT scan)

5. Patient Care (hospital beds, wheelchairs)

6. Rehabilitation Robotics (physical therapy)

7. Medical Instrumentation (surgical tools)

3) Aerospace and Defense:

1. Aircraft Control Surfaces (flaps, ailerons)

2. Satellite Deployment and Maneuvering

3. Space Exploration (robotic arms, rover systems)

4. Missile Guidance Systems

5. Unmanned Aerial Vehicles (UAVs)

6. Autonomous Underwater Vehicles (AUVs)

4) Consumer and Service Robotics:

1. Vacuum Cleaning Robots (Roomba)

2. Lawn Mowing Robots (Husqvarna)

3. Pool Cleaning Robots

4. Home Security Systems

5. Personal Assistants (robotic arms)

6. Autonomous Wheelchairs

5) Automotive Applications:

1. Anti-Lock Braking Systems (ABS)

2. Electronic Stability Control (ESC)

3. Power Steering

4. Automatic Transmission

5. Engine Control Units (ECU)

6. Autonomous Vehicles (self-driving cars)

6) Agricultural Applications:

1. Autonomous Farm Equipment

2. Crop Monitoring and Management

3. Livestock Monitoring

4. Harvesting and Pruning Robots

5. Green house Automation

ACTUATORS IN JOINTS AND WHEELS

Joint-Specific Actuators

1. Shoulder Joint: Rotary actuator (multi-axis movement)

2. Elbow Joint: Linear actuator (extension/retraction)

3. Wrist Joint: Rotary actuator (flexion/extension)

4. Finger Joints: Linear actuators (flexion/extension)

5. Hip Joint: Rotary actuator (multi-axis movement)

6. Knee Joint: Linear actuator (extension/retraction)

Examples:

- Humanoid robots

- Industrial robots

- Prosthetic limbs

- Exoskeletons

Wheel-Specific Actuators

1. Steering Wheels: Rotary actuators (direction control)

2. Drive Wheels: Rotary actuators (motion control)

3. Caster Wheels: Linear actuators (direction control)

Examples:

- Autonomous vehicles

- Industrial robots

- Agricultural equipment

- Medical devices (e.g., wheelchairs)

CONTROLLER OF A ROBOTIC SYSTEM

A controller in a robotic system is a critical component responsible for managing and regulating the robot's movements, actions, and interactions. They are devices or systems responsible for commanding and coordinating the actions and behaviours of robots.

DAILY LIFE EXAMPLES

a) **Air conditioner**

System Components:

1. Sensor (Thermostat): Measures room temperature.
2. Controller (Microcontroller/Thermostat): Processes sensor data and controls AC.

3. Actuator (Compressor, Fan Motor): Adjusts cooling and airflow.

4. Feedback Loop: Room temperature sensor provides feedback to controller.

b) Regulation of fan

System Components:

1. Sensor (Temperature Sensor): Measures room temperature.

2. Controller (Thermostat/Microcontroller): Processes sensor data and controls fan.

3. Actuator (Fan Motor): Adjusts fan speed.

4. Feedback Loop: Room temperature sensor provides feedback to controller.

c) Refrigerator

System Components:

1. Sensor (Temperature Sensor): Measures refrigerator temperature.

2. Controller (Thermostat/Microcontroller): Processes sensor data and controls compressor.

3. Actuator (Compressor, Fan Motor): Adjusts cooling and airflow.

4. Feedback Loop: Temperature sensor provides feedback to controller.

FUNCTIONS AND WORKING OF CONTROL SYSTEM

Diagram from chapter 2, class 9, pgno 50

Functions of Control Systems:

1. Regulation: Maintain desired output despite disturbances.

2. Stabilization: Prevent oscillations and instability.

3. Optimization: Achieve optimal performance and efficiency.

4. Protection: Safeguard against faults, errors, and hazards.

5. Automation: Perform tasks without human intervention.

Working of Control Systems:

1. Sensing: Measure process variables (temperature, pressure, flow rate, etc.).

2. Comparison: Compare measured values to setpoints (desired values).

3. Computation: Calculate control actions based on errors (deviations).

4. Actuation: Implement control actions (adjust valves, motors, etc.).

5. Feedback: Monitor process variables and adjust control actions.

TYPES OF CONTROL SYSTEMS

a) Manual control systems
b) Automatic control systems

Manual control systems

A manual control system is a type of control system where human operators manually regulate and monitor processes.

How it Works

1. Operators monitor sensors and indicators.

2. Operators adjust actuators based on process variable values.

3. Process variables change in response to actuator adjustments.

4. Operators continuously monitor and adjust.

Examples

1. Manual temperature control in homes

2. Manual gear shifting in vehicles

Advantages

1. Low cost

2. Simple implementation

3. Easy maintenance

4. Flexibility

5. Human judgment and expertise

Disadvantages

1. Limited accuracy

2. Prone to human error

3. Slow response times

4. Labor-intensive

5. Limited scalability

Automatic control systems

An automatic control system is a type of control system that uses electronic or mechanical devices to regulate and monitor processes without human intervention.

How it Works

1. Sensors measure process variables.

2. Controllers process sensor data and calculate control actions.

3. Actuators implement control actions.

4. Feedback loop adjusts control actions based on sensor feedback.

Examples

1. Speed control in motor drives.

2. Pressure control in hydraulic systems.

3. Flow control in pipeline systems.

Advantages

1. High accuracy and precision.

2. Fast response times.

3. Increased efficiency and productivity.

4. Reduced human error and fatigue.

5. Improved reliability and stability.

Disadvantages

1. Higher initial cost.

2. Complexity requires specialized expertise.

3. Maintenance and repair challenges.

4. Dependence on electronics and software.

BLOCK DIAGRAM

Pgno 91

INTEGRATING SENSORS, ACTUATORS AND CONTROLLER IN A ROBOTIC SYSTEM

Integrating sensors, actuators, and controllers is crucial for building a functional robotic system.

Integration Steps

1. Define system requirements and goals.

2. Choose sensors and actuators suitable for the application.

3. Design and implement the control algorithm.

4. Integrate sensors, actuators, and controller using communication protocols

5. Test and calibrate the system

Explain in simple way how angular position is measured by a sensor of a robotic arm is controlled by driving the revolute joint using a motor?

Components:

1. Robotic arm

2. Revolute joint (moves in a circular motion)

3. Motor (drives the joint)

4. Sensor (measures angular position)

5. Controller (processes sensor data and controls motor)

How it works:

1. The sensor measures the current angle of the joint.

2. The controller compares this angle to the desired angle (setpoint).

3. If the angles don't match, the controller sends a signal to the motor.

4. The motor rotates the joint to match the desired angle.

5. The sensor continuously measures the new angle and sends it to the controller.

6. The controller adjusts the motor signal to maintain the desired angle.